A Graph Representation of DJ Tracklists

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Abstract

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Glossary

Term	Meaning
BPM	Synonyms: tempo, speed.
Combo	A connection between two tracks. Synonyms: match. A combo is identified in the network as an edge. Combos are associated with a metadata vector containing subjective data (e.g. rating and frequency of inclusion in walks).
Edge	...
Fingerprint	Synonyms: style. Quantitative (statistical) properties of a set of walks associated with a DJ's tracklist. (perhaps choose different name)
Key	Synonyms: pitch, scale, tone.
Node	...
Node parameters	...
Track	A musical unit. Synonyms: song, record, tune. A track is defined by a metadata vector containing bookkeeping information (e.g. name and release date) and musical information (e.g. key and tempo). A track is identified in the network as a node. Quantitative metadata is identified as values of node parameters.
Track set	An unordered number of tracks. Synonyms: playlist.
Track vector	An ordered track set. Synonyms: tracklist, DJ set (list), track sequence. Subsequent elements in a track vector correspond to tracks that are played in chronological order. A track vector is identified in the network as a walk.
Walk	...

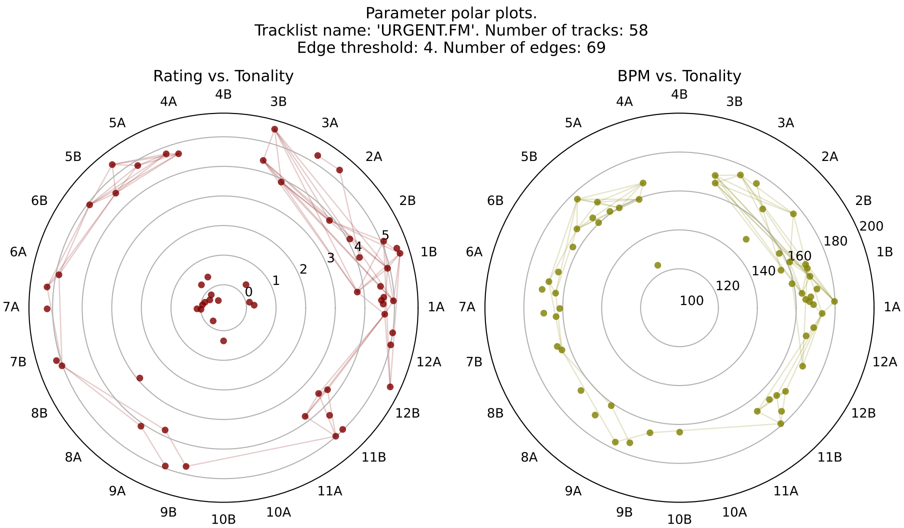

Figure 1: Example of two polar plots showing the distribution of the musical characteristics. Nodes between edges suggests the possibility of a combo. The threshold here is 4, meaning that a combo is only suggested if all 4 couples of parameter values match. Concretely: the genres must be the same, the BPM must be maximally 10% apart, the key must be adjacent in the Camelot wheel, and the rating must maximally differ by one star.

1 Context

2 Examples

3 Synthesis of Track Set Graphs and Track Vector Walks

3.1 Track Set Graphs

An unordered set of tracks can be represented as nodes in a graph. The edges between nodes represent some relationship between tracks and (hence) suggest that these tracks may form a good combo.

In the simplest case, edges are either there or they aren't. In the more elaborate case, edges are associated with weights (a combo suggestion may be strong or may be weak). In the former case, which we will call binary edge

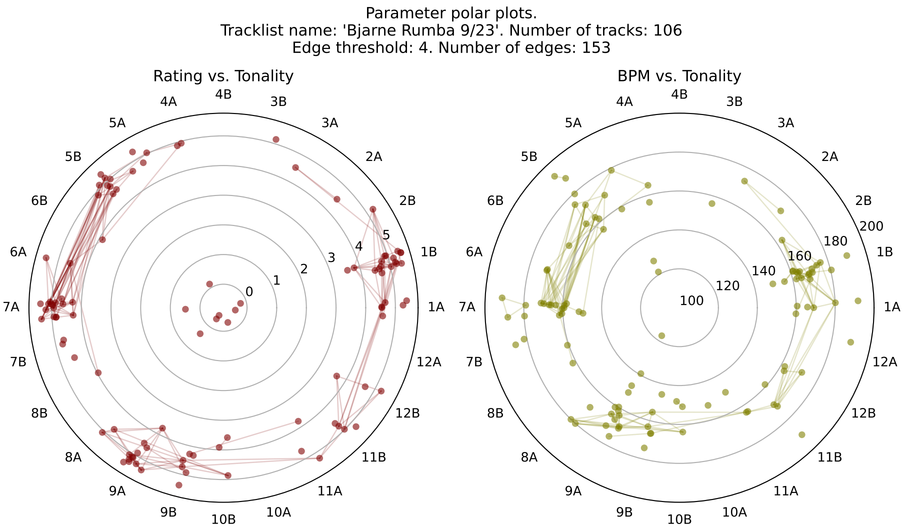

Figure 2: Same as previous image, but with a different playlist.

formation, some criteria must be met based on the parameter values of both tracks. A naive example of such criteria is shown in Tab. 1

Table 1: Example of node parameters and their required ranges that are considered when deciding whether two nodes (tracks) ought to be linked with an edge (suggested combo). Variation within a particular choice is possible: adding properties and altering required value ranges. With a particular choice, variation of tolerance is possible: should strictly all requirements be met, or is a certain fraction already tolerated?

Parameter	**Required value range**	**Example of match**
BPM	±10%	130 BPM vs. 120 BPM
Key	Adjacent in Camelot wheel	9A vs. 9B
Year	±5 years	1998 vs. 1995

3.2 Track Vector Walks

4 Analysing Existing DJ Sets

A suggested path:

- Listing the most common types of walks on networks
- Listing qualitative classifications and quantitative metrics used to characterise walks.
- Identify relationships between network types and sizes on the one hand, and walk characteristics on the other hand.
- Download track vectors of established DJs. Tabulate all metadata.
- For each DJ, append all played tracks to a single track set.
- Use a selection of metrics to quantify track vectors within the track set. Apply relevant statistics to the resulting set of quantities. These statistics represent a DJ's 'fingerprint'.
- Plot and compare various DJ fingerprints. Is there any significant difference?
- Possible: apply predictive modelling techniques (such as principal component analysis) to select for fingerprint properties that perform best at distinguishing DJs.
- Assign subjective and objective properties to DJs. Are there correlations between DJ properies and DJ fingerprints?

One bottleneck will of course be – as usual – the difficulty of acquiring good data and optimising data processing efficiency.

5 Applying DJ Set Characteristics to Tracklist Synthesis

General idea in technical terms. A track vector induces some walk in a network. Some established metrics describe such walks quantitatively. A DJ is associated with an (possibly ordered) set of walks, allowing for the extraction of a 'spectrum of walks'. Characteristics of this spectrum are a DJ's fingerprint. These characteristics can be used as weights during preferential attachment of a (new) track set.

General idea in human terms. Within a DJs track set, even of moderate size, the number of possible tracklists is virtually endless. Rather than building one's own tracklist, one can have the software suggest one. It could either suggest an entire tracklist, or suggest a single next track. The way that the software determines this tracklist can obviously be fully random, or based on objective musical characteristics. However, its choice an also be inspired by (and similar to) how a particular DJ would be inclined to choose tracks within this track set.

Ideally, a user can influence the importance that is given to each mechanism controlling the suggestions. Do you want a bit more randomness? Do you want to really stress the importance of remaining at similar energy levels? Do you want to make track selections exactly how Diplo would choose it, or rather a mix between how DJ Hype and CJ Bolland compile their tracklists? This should be as easy as adjusting a percentage in the settings.

6 Additional Features

6.1 Node, Edge and Walk Parameters

Objective parameters:

- Presence of lyrics (percentage) (determined with AI)
- Release date
- Number of listens on Spotify (logarithmic scale)
- Some metric in the space of the embedded vectors, using deep neural encodings of music with e.g. `music2vec`[1]
- A similar distance may be defined for tonal harmony [1]

Subjective parameters:

- Feedback from users on edges, e.g. negative advice despite objective match.
- Overall rating (percentage)
- More specific rating. These could also be some gimmick, some inside joke, e.g. have users give a percentage to "would play at a funeral".

[1] https://medium.com/@rajatheb/music2vec-generating-vector-embedding-for-genre-classification-task-411187a2082

- A walk imported from an actual DJ tracklist could also contain information on when the transition is made, how sudden the transition was, ... This would be very valuable information in characterising the fingerprint of a DJ's style.

6.2 User Experience Features

View of the network:

- Show network in two dimensions, choosing which dimensions are displayed.
- A third dimension may be expressed as a node color.
- When hovering over a node, information should appear.
- DJ should have the option to 'randomise' connections, which performs a (preferential) reattachment of nodes.
- Edges can take different colours or shapes. For example (1) no edge between nodes means 'combo unexplored'. Red ...
- It should be very easy to highlight a walk through the network as one DJ set.
- It should be very easy to play with this. Adding a track at the end? Continue the walk. Adding a track in the middle of tracks X and Y? Drag the middle of the edge between X and Y to the track that is being squeezed in.
- Include option in settings: edges should be able to show directionality.
- Show network in three dimensions in Virtual Reality.
- ...

7 Discussion and Conclusion

Problems

- There will be many more nodes and edges than currently documented walks (tracklists). That is to say: the data set is probably too small to make quantitative statistical statements on tracklist styles.

- ...

Scientific interest:

- Note: this is only related to network analysis, not (immediately) to network synthesis.
- What is the character of the network expressed by the adjacency matrix? Small-world network? Clustered network? Can this be determined by means of network automata?
- What is the character of typical walks through this network? Random walk, tourist walk?
- What variation is found in network walks and how is this correlated to character parameters of the DJ.
- What is the effect of network size (proportional to number of included DJ sets) on network characteristics?
- Is walk characterisation robust to adding nodes to the track set?
- ...

Creative interest:

- This is a new approach to expressing connections between tracks. Nonlinear in time.
- Those connections allow for the "story-telling" aspect that makes a DJ selections superior to a random tracklist order.
-
- Resulting graphs are by themselves a piece of art.
- You could generate a tracklist like some DJ would, but from a track set containing music that that DJ would never play. How about selecting samba music like Shpongle would? Note: this hinges on the supposed unique 'fingerprint' of DJs, which is not at all certain to be a distinguishing characteristic.
- ...

Entrepreneurial interest:

- This allows for mapping their music collection in a way that is much more informative than the linear presentation in conventional DJ software.
- It can be integrated in conventional DJ software quite easily, as an additional feature, which increases the algorithm's projected market value.
- This allows for DJs to learn from established DJs. This implies that any product that integrates this algorithm would immediately become a learning tool, aside from being a organisational tool. This implies that more DJs would recognise its added value.
- This – and I continue to stress the market value here – also allows for artificial intelligence to learn how to behave as creatively (as human?) as humans.
- Note that we do not provide the AI that can interpret these human patterns, but (perhaps even importantly) we provide that data used for effectively training this AI.
- This means: high market value for established (and impossible-to-compete-with) AI companies, because they would equally realise the demand for 'artificial DJs'.
- With this AI (which, again, we won't produce but merely 'feed'), any low-budget café could (1) select or generate a tracklist, and (2) have it mixed in the way that DJ X would mix it. All without ever copying human DJ sets, and without playing the same set twice.
- ...

A More ideas

1. The default setting is: key as angle, BPM as radius, colour coding as rating
2. It should be possible to show which keys are minor and which are major, either by toggling it in the settings, or by default with a very subtle touch. I prefer the latter (keep it simple), e.g. by making the edge of the 'Camelot wheel' a bit bigger.
3. We should have a good policy on what to do with missing data.

4. Visually, we could have the alpha of nodes or edges depend on which track we have currently selected. That is to say: if you're currently looking for a match with track X, you do not need to see the *entire* network. Only the nodes that are separated by a certain maximum degree, with other nodes disappearing in the background.

5. When making a track vector, the user should be able to see the network structure, but also see the linear list with metadata on artist and track name.

6. Make a widget in `matplotlib` to control the combo criteria, which should interactively change the visual output. I see two major (possibly complementary) approaches:

    - Change the combo criterion (e.g. maximal percentage of BPM change) for each of the parameters. The resulting mesh of edges is Boolean: either it passes the check, or it doesn't.
    - Change the weights of the criteria, expressing how much you care about certain parameter values matching. E.g. whether the keys match is not important, but they *really* both must have the same energy level. The resulting strength of the match can be visually expressed as alpha or width of the edge.

7. Visually, it should be possible (perhaps as recommended first step) to look at your network of tracks without imposing structure, i.e. without putting the nodes on a polar plot. This allows the user to define criteria that result in a 'healthy' network, that is not cluttered but at the same time provides enough suggestions.

8. Just for the heck of it, make a classification neural network that can predict the genre based on some characteristics. Determine which characteristics are most important (principal component analysis). There is good data to play around with.

References

[1] W. B. de Haas, F. Wiering, and R. C. Veltkamp, "A geometrical distance measure for determining the similarity of musical harmony," *International Journal of Multimedia Information Retrieval*, vol. 2, pp. 189–202, Sep 2013.